

Лабораторная работа №01

Моделирование сетей передачи данных

бахи сиди али темассини

2026-09-12

Содержание I

- 1 Информация
- 2 Цель работы
- 3 Выполнение лабораторной работы
- 4 Заключение

Раздел 1

Информация

- бахи сиди али темассини
- Российский университет дружбы народов

Раздел 2

Цель работы

Цель работы

- Изучить основные возможности среды Mininet.
- Создать простую сетевую топологию.
- Настроить IP-адреса хостов.
- Проверить связность между узлами.
- Освоить работу с графическим редактором MiniEdit.

Раздел 3

Выполнение лабораторной работы

Запуск виртуальной машины Mininet

- Выполнен вход под пользователем mininet.
- Командой `ifconfig` проверены сетевые интерфейсы.
- Интерфейс `eth0`: 192.168.200.129.
- Проверен локальный интерфейс `lo`.

```
Ubuntu 20.04.1 LTS mininet-vm tty1

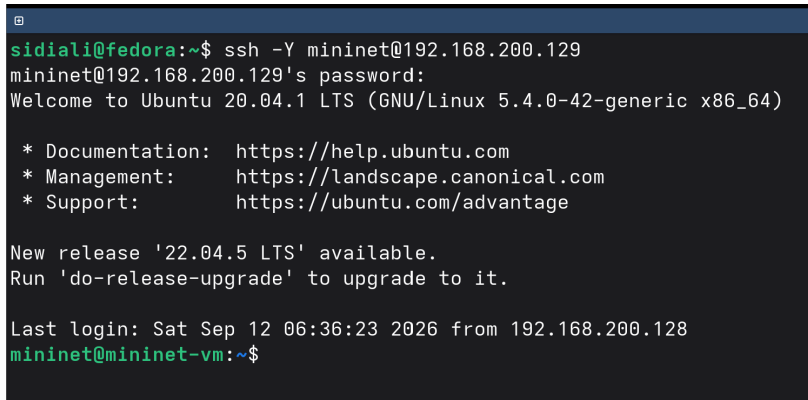
mininet-vm login: mininet
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Last login: Wed Feb 10 21:03:31 PST 2021 on ttyS0
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.200.129 netmask 255.255.255.0 broadcast 192.168.200.255
    ether 00:0c:29:66:e2:dd txqueuelen 1000 (Ethernet)
    RX packets 134 bytes 25030 (25.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 122 bytes 11711 (11.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Подключение по SSH

- Выполнено подключение с перенаправлением X11: `ssh -Y mininet@192.168.200.129`.
- Соединение с виртуальной машиной успешно установлено.

A terminal window showing the process of connecting to a virtual machine named 'mininet' via SSH. The user 'sidiali' on a 'fedora' machine runs the command 'ssh -Y mininet@192.168.200.129'. The terminal shows the password prompt, a welcome message for Ubuntu 20.04.1 LTS, system information, links to documentation, management, and support, a notification about a new release, and the last login time. Finally, the prompt changes to 'mininet@mininet-vm:~\$', indicating a successful connection.

```
sidiali@fedora:~$ ssh -Y mininet@192.168.200.129
mininet@192.168.200.129's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Sep 12 06:36:23 2026 from 192.168.200.128
mininet@mininet-vm:~$
```

Рисунок 2: Подключение к виртуальной машине Mininet по протоколу SSH

Настройка SSH-ключа

- SSH-ключ скопирован командой `ssh-copy-id`.
- Повторное подключение выполнено с параметром `-Y`.
- Подключение выполнено без запроса пароля.

```
mininet@mininet-vm:~$ logout
Connection to 192.168.200.129 closed.
sidiali@fedora:~$ ssh-copy-id mininet@192.168.200.129
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
/usr/bin/ssh-copy-id: ERROR: No identities found
sidiali@fedora:~$ ssh-copy-id mininet@192.168.200.129
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
/usr/bin/ssh-copy-id: ERROR: No identities found
sidiali@fedora:~$ ssh -Y mininet@192.168.200.129
mininet@192.168.200.129's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

Проверка сетевой конфигурации

- Повторно выполнена команда `ifconfig`.
- `eth0: 192.168.200.129`.
- `lo: 127.0.0.1`.

```
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.200.129  netmask 255.255.255.0  broadcast 192.168.200.255
    ether 00:0c:29:66:e2:dd  txqueuelen 1000  (Ethernet)
    RX packets 881  bytes 90472 (90.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 564  bytes 60044 (60.0 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 28  bytes 2824 (2.8 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 28  bytes 2824 (2.8 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

mininet@mininet-vm:~$
```

Рисунок 4: Проверка текущей конфигурации сетевых интерфейсов

Настройка интерфейса eth1

- Запущен DHCP-клиент для eth1.
- Интерфейсу назначен адрес 192.168.100.128.
- Подтверждена работа двух сетевых интерфейсов.

```
mininet@mininet-vm:~$ sudo dhclient -v eth1
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth1/00:0c:29:66:e2:e7
Sending on   LPF/eth1/00:0c:29:66:e2:e7
Sending on   Socket/fallback
DHCPREQUEST for 192.168.12.128 on eth1 to 255.255.255.255 port 67 (xid=0x3a9ea0c0)
DHCPACK of 192.168.12.128 from 192.168.12.254 (xid=0xc0a09e3a)
RTNETLINK answers: File exists
bound to 192.168.12.128 -- renewal in 681 seconds.
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.200.129 netmask 255.255.255.0 broadcast 192.168.200.255
    ether 00:0c:29:66:e2:dd txqueuelen 1000 (Ethernet)
    RX packets 1087 bytes 108320 (108.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 700 bytes 76518 (76.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.12.128 netmask 255.255.255.0 broadcast 192.168.12.255
    ether 00:0c:29:66:e2:e7 txqueuelen 1000 (Ethernet)
    RX packets 10 bytes 2108 (2.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
```

Установка Midnight Commander

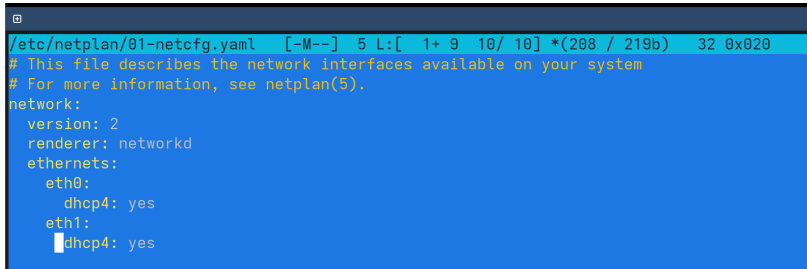
- Установлен файловый менеджер mc.
- Установка выполнена командой `sudo apt install mc`.
- Необходимые зависимости установлены успешно.

```
mininet@mininet-vm:~$ sudo apt install mc
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libssh2-1 mc-data unzip
Suggested packages:
  arj catdvi | texlive-binaries dbview djvulibre-bin epub-utils genisoimage gv imagemagick libaspell-dev links | w3m | lynx odt2txt poppler-utils python python-boto python-tz xpdf | pdf-viewer zip
The following NEW packages will be installed:
  libssh2-1 mc mc-data unzip
0 upgraded, 4 newly installed, 0 to remove and 84 not upgraded.
Need to get 1,986 kB of archives.
After this operation, 8,587 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 libssh2-1 amd64 1.8.0-2.1build1 [75.4 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 mc-data all 3:4.8.24-2ubuntu1 [1,285 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 mc amd64 3:4.8.24-2ubuntu1 [477 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu focal/main amd64 unzip amd64 6.0-25ubuntu1 [169 kB]
Fetched 1,986 kB in 3s (641 kB/s)
Selecting previously unselected package libssh2-1:amd64.
(Reading database ... 181729 files and directories currently installed.)
Preparing to unpack .../libssh2-1.1.0-2.1build1_amd64.deb ...
Unpacking libssh2-1:amd64 (1.8.0-2.1build1) ...
Selecting previously unselected package mc-data.
Preparing to unpack .../mc-data_3k3a4.8.24-2ubuntu1_all.deb ...
Unpacking mc-data (3:4.8.24-2ubuntu1) ...
Selecting previously unselected package mc.
Preparing to unpack .../mc_3k3a4.8.24-2ubuntu1_amd64.deb ...
Unpacking mc (3:4.8.24-2ubuntu1) ...
Selecting previously unselected package unzip.
Preparing to unpack .../unzip_6.0-25ubuntu1_amd64.deb ...
Unpacking unzip (6.0-25ubuntu1) ...
Setting up unzip (6.0-25ubuntu1) ...
Setting up mc-data (3:4.8.24-2ubuntu1) ...
Setting up libssh2-1:amd64 (1.8.0-2.1build1) ...
Setting up mc (3:4.8.24-2ubuntu1) ...
Processing triggers for man-db (2.8.1-1) ...
Processing triggers for alse-support (3.04ubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.31-0ubuntu9) ...
```

Рисунок 6: Установка файлового менеджера Midnight Commander в виртуальной машине

Настройка Netplan

- Изменён файл `/etc/netplan/01-netcfg.yaml`.
- Для `eth0` и `eth1` включена автоматическая настройка IPv4 по DHCP.



```
/etc/netplan/01-netcfg.yaml  [-M--]  5 L:[  1+ 9  10/ 10] *(208 / 219b)  32 0x020
# This file describes the network interfaces available on your system
# For more information, see netplan(5).
network:
  version: 2
  renderer: networkd
  ethernets:
    eth0:
      dhcp4: yes
    eth1:
      dhcp4: yes
```

Рисунок 7: Настройка DHCP для сетевых интерфейсов `eth0` и `eth1`

- Исходный каталог переименован в `mininet.orig`.
- Исходный код Mininet клонирован повторно.
- Выполнена установка командой `sudo make install`.
- Версия Mininet: 2.3.1b4.

```

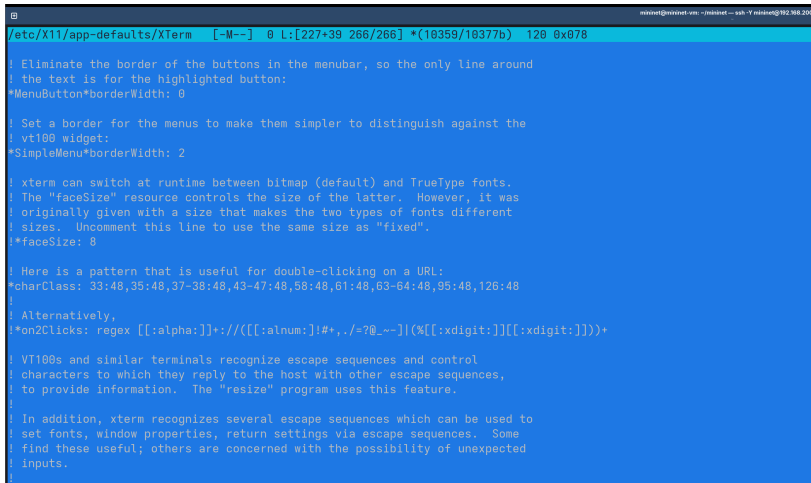
mininet@mininet-vm:~$ mv ~/mininet ~/mininet.orig
mininet@mininet-vm:~$ cd ~
mininet@mininet-vm:~$ git clone https://github.com/mininet/mininet.git
Cloning into 'mininet'...
remote: Enumerating objects: 10388, done.
remote: Counting objects: 100% (136/136), done.
remote: Compressing objects: 100% (64/64), done.
remote: Total 10388 (delta 100), reused 72 (delta 72), pack-reused 10252 (from 2)
Receiving objects: 100% (10388/10388), 3.36 MiB | 1.35 MiB/s, done.
Resolving deltas: 100% (8909/8909), done.
mininet@mininet-vm:~$ cd ~/mininet
mininet@mininet-vm:~/mininet$ ls
bin  CONTRIBUTORS  custom  debian  doc  examples  INSTALL  LICENSE  Makefile  mininet  mnexec.c  README.md  setup.py  util
mininet@mininet-vm:~/mininet$ sudo make install
cc -Wall -Wextra \
-DVERSION="\`PYTHONPATH=. python -B bin/mn --version 2>&1\`" mnexec.c -o mnexec
install -D mnexec /usr/bin/mnexec
PYTHONPATH=. help2man -N -n "create a Mininet network." \
--no-discard-stderr "python -B bin/mn" -o mn.1
help2man -N -n "execution utility for Mininet." \
-h "-" -v "-" --no-discard-stderr ./mnexec -o mnexec.1
install -D -t /usr/share/man/man1 mn.1 mnexec.1
python -m pip uninstall -y mininet || true
Found existing installation: mininet 2.3.0
Uninstalling mininet-2.3.0:
  Successfully uninstalled mininet-2.3.0
python -m pip install .
Processing /home/mininet/mininet
Requirement already satisfied: setuptools in /usr/lib/python3/dist-packages (from mininet==2.3.1b4) (45.2.0)
Building wheels for collected packages: mininet
  Building wheel for mininet (setup.py) ... done
  Created wheel for mininet: filename=mininet-2.3.1b4-py3-none-any.whl size=160942 sha256=3846107a1ed882d506fc79ebad6a09a83dbd67dc89823b99491c5ed18de49258
  Stored in directory: /tmp/pip-ephem-wheel-cache-e9p3a1n/wheels/cd/7d/a7/aaf1b3eaff31efd6ba4e2ea6c9690a717bdf739db6cfe8d45
Successfully built mininet
Installing collected packages: mininet
Successfully installed mininet-2.3.1b4
mininet@mininet-vm:~/mininet$ mn --version
2.3.1b4
mininet@mininet-vm:~/mininet$

```

Рисунок 8: Установка Mininet из исходного кода и проверка версии

Настройка терминала xterm

- В конфигурации xterm задан размер шрифта.
- Установлен параметр xterm*faceSize: 12.



```
mininet@mininet-vm: ~/mininet -- ssh -Y mininet@192.168.200.10
/etc/X11/app-defaults/XTerm  [-M--]  0 L:[227+39 266/266] *(10359/10377b) 120 0x078

! Eliminate the border of the buttons in the menubar, so the only line around
! the text is for the highlighted button:
*MenuBar*borderWidth: 0

! Set a border for the menus to make them simpler to distinguish against the
! vt100 widget:
*SimpleMenu*borderWidth: 2

! xterm can switch at runtime between bitmap (default) and TrueType fonts.
! The "faceSize" resource controls the size of the latter.  However, it was
! originally given with a size that makes the two types of fonts different
! sizes.  Uncomment this line to use the same size as "fixed".
!*faceSize: 8

! Here is a pattern that is useful for double-clicking on a URL:
*charClass: 33:48,35:48,37-38:48,43-47:48,58:48,61:48,63-64:48,95:48,126:48
!
! Alternatively,
!*on2Clicks: regex [[[:alpha:]]+//([[:alnum:]]!#+,./=?@_~)|(%[[:xdigit:]][:xdigit:]]))+

! VT100s and similar terminals recognize escape sequences and control
! characters to which they reply to the host with other escape sequences,
! to provide information.  The "resize" program uses this feature.
!
! In addition, xterm recognizes several escape sequences which can be used to
! set fonts, window properties, return settings via escape sequences.  Some
! find these useful; others are concerned with the possibility of unexpected
! inputs.
```

Настройка X11-аутентификации

- Получено значение MIT-MAGIC-COOKIE-1.
- Значение добавлено для пользователя root.
- Настройка проверена командой `xauth list $DISPLAY`.

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:12 MIT-MAGIC-COOKIE-1 1e698e64bc4a368db176c3601ee8d8c5
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth list
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 e6ba0b8484b25512b5adb71ca45d6a9c
root@mininet-vm:~# xauth add mininet-vm/unix:12 MIT-MAGIC-COOKIE-1 1e698e64bc4a368db176c3601ee8d8c5
root@mininet-vm:~# xauth list $DISPLAY
mininet-vm/unix:12 MIT-MAGIC-COOKIE-1 1e698e64bc4a368db176c3601ee8d8c5
root@mininet-vm:~# logout
mininet@mininet-vm:~$
```

Рисунок 10: Настройка X11-аутентификации для пользователя root

Запуск стандартной топологии Mininet

- Mininet запущен командой `sudo mn`.
- Созданы хосты `h1` и `h2`.
- Созданы коммутатор `s1` и контроллер `c0`.
- Командой `help` просмотрены доступные команды.

```

mininet@mininet-vm:~$ sudo mn
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> help

Documented commands (type help <topic>):
=====
EOF      gterm  iperfudp  nodes      pingpair    py      switch  xterm
dpctl    help   link      noecho     pingpairfull  quit    time
dump     intfz  links     pingall    ports       sh      wait
exit     iperf  net       pingallfull  px          source  x

You may also send a command to a node using:
  <node> command {args}
For example:
  mininet> h1 ifconfig

The interpreter automatically substitutes IP addresses
for node names when a node is the first arg, so commands
like
  mininet> h2 ping h3
should work.

Some character-oriented interactive commands require
noecho:
  mininet> noecho h2 vi foo.py

```

Рисунок 11: Запуск стандартной топологии и просмотр команд Mininet

Проверка узлов и интерфейсов

- Командами `nodes` и `net` проверена топология.
- Хост `h1`: `10.0.0.1`.
- Хост `h2`: `10.0.0.2`.

```
mininet> nodes
available nodes are:
c0 h1 h2 s1
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 62:47:47:d5:04:79 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether d8:d3:88:88:0a:c5 txqueuelen 1000 (Ethernet)
```

- Выполнен `ping` между `h1` и `h2`.
- Передано и получено 5 пакетов.
- Потери пакетов: 0%.
- RTT: 0.048/0.369/1.467/0.552 мс.
- Работа Mininet завершена командой `exit`.

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.63 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.457 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.031 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.035 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.040 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.077 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.035 ms
^C
--- 10.0.0.2 ping statistics ---
24 packets transmitted, 24 received, 0% packet loss, time 23529ms
rtt min/avg/max/mdev = 0.031/0.171/2.630/0.519 ms
mininet> exit
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 273.827 seconds
mininet@mininet-vm:~$
```

Запуск MiniEdit

- Запущен графический редактор MiniEdit.
- Открыто рабочее окно с пустой топологией.

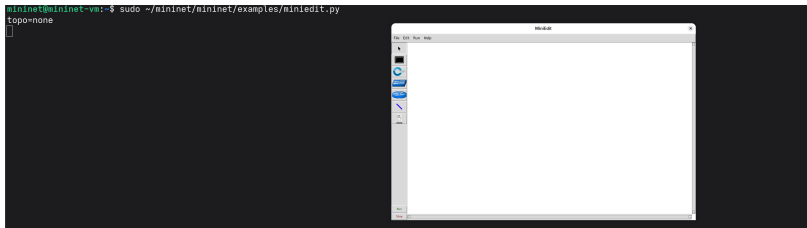


Рисунок 14: Запуск графического редактора топологий MiniEdit

Создание топологии в MiniEdit

- Созданы два хоста: h1 и h2.
- Добавлен коммутатор s1.
- Хосты соединены через коммутатор.

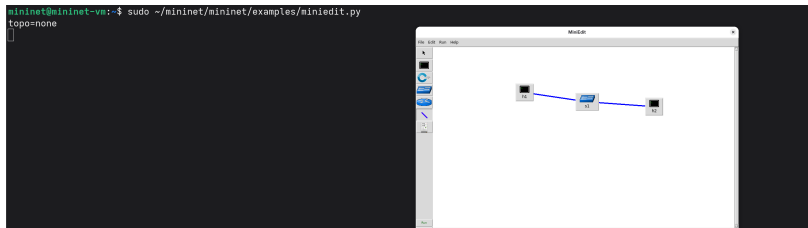


Рисунок 15: Топология сети из двух хостов и одного коммутатора в MiniEdit

Настройка IP-адреса h1

- Открыты свойства хоста h1.
- Задан IP-адрес 10.0.0.1/8.

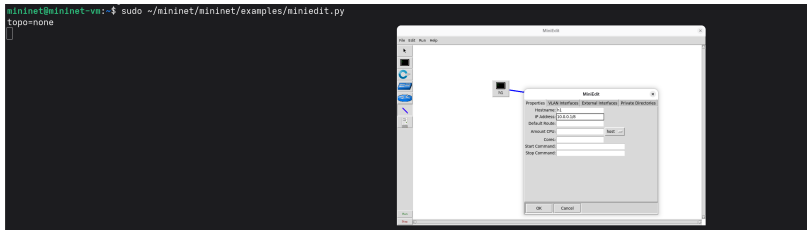


Рисунок 16: Настройка IP-адреса хоста h1 в MiniEdit

Настройка IP-адреса h2

- Открыты свойства хоста h2.
- Задан IP-адрес 10.0.0.2/8.

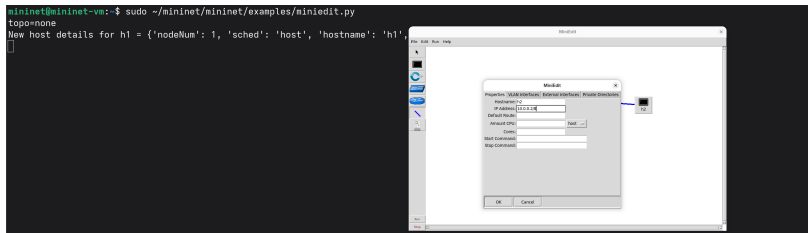


Рисунок 17: Настройка IP-адреса хоста h2 в MiniEdit

- Запущена созданная топология.
- Проверены адреса командой `ifconfig`.
- Выполнен взаимный `ping` между `h1` и `h2`.
- Пакеты переданы без потерь.

```
mininet@mininet-vm:~$ sudo ~/mininet/mininet/examples/miniedit.py
topo=none
New host details for h1 = ('nodeNum': 1, 'schema': 'h1', 'ip': '10.0.0.1', 'mac': '00:00:00:00:00:00', 'mtu': 1500)
New host details for h2 = ('nodeNum': 2, 'schema': 'h2', 'ip': '10.0.0.2', 'mac': '00:00:00:00:00:00', 'mtu': 1500)
New host details for h1 = ('nodeNum': 1, 'schema': 'h1', 'ip': '10.0.0.1', 'mac': '00:00:00:00:00:00', 'mtu': 1500)
Getting Hosts and Switches.
Getting controller selection:ref
Getting controller selection:ref
Getting Links.
*** Configuring hosts
h1 h2
**** Starting 2 controllers
c0 c1
**** Starting 1 switches
s1
No NetFlow targets specified.
No sflow targets specified.
[ ]
```

Terminal window "Host h1" output:

```
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 02:00:00:00:00:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

to: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 762 bytes 233124 (233.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 762 bytes 233124 (233.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.478 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.468 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.455 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.434 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.477 ms
```

Terminal window "Host h2" output:

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 02:00:00:00:00:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

to: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1929 bytes 243962 (243.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1929 bytes 243962 (243.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=0.349 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.035 ms
... 10.0.0.1 ping statistics ...
2 packets transmitted, 2 received, 0% packet loss, time 1028ms
```

Рисунок 18: Проверка IP-адресов и связности хостов созданной топологии

Удаление IP-адресов хостов h1

- Удалён вручную заданный IP-адрес хоста h1.
- Поле IP Address оставлено пустым.

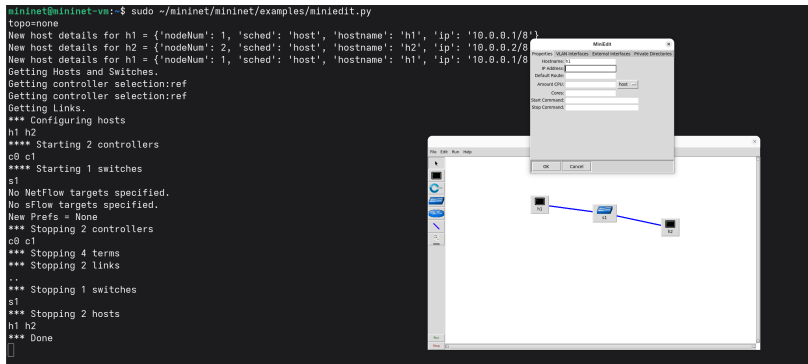


Рисунок 19: Удаление вручную заданного IP-адреса хоста h1

Удаление IP-адресов хостов h2

- Удалён вручную заданный IP-адрес хоста h2.
- Поле IP Address оставлено пустым.

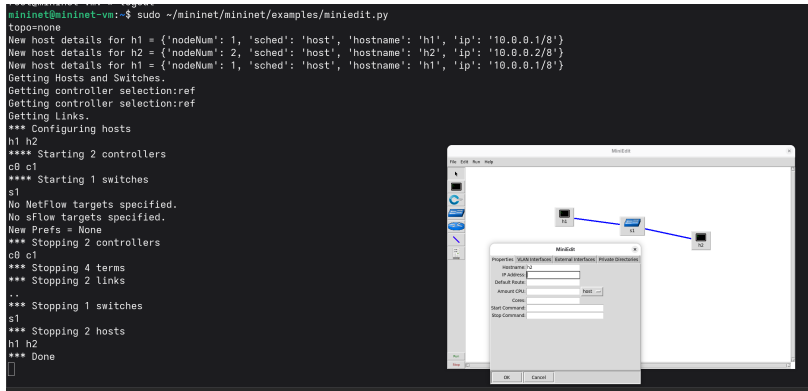


Рисунок 20: Удаление вручную заданного IP-адреса хоста h2

Настройка IP Base

- В настройках MiniEdit изменён параметр IP Base.
- Задана базовая сеть 15.0.0.0/8.
- Адреса хостов назначаются автоматически.

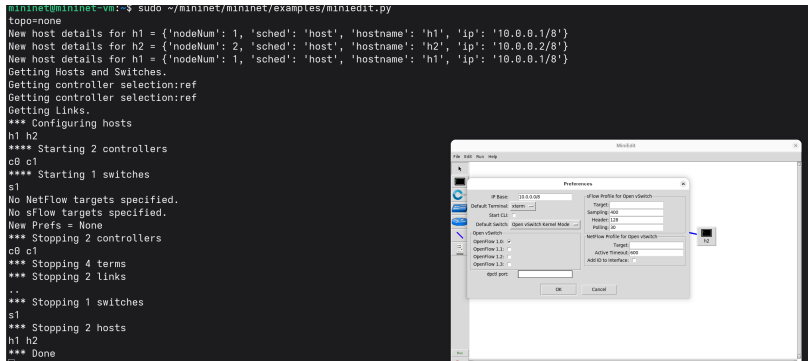


Рисунок 21: Настройка базовой IP-сети 15.0.0.0/8 в MiniEdit

Проверка автоматической адресации

- Топология запущена повторно.
- Хост h1 получил адрес 15.0.0.1.
- Хост h2 получил адрес 15.0.0.2.
- Маска сети: 255.0.0.0.
- IP Base обеспечивает автоматическое назначение адресов.

```

topo=None
New host details for h1 = {'nodeNum': 1, 'sched': 'host', 'hostname': 'h1', 'ip': '10.0.0.1/8'}
New host details for h2 = {'nodeNum': 2, 'sched': 'host', 'hostname': 'h2', 'ip': '10.0.0.2/8'}
New host details for h1 = {'nodeNum': 1, 'sched': 'host', 'hostname': 'h1', 'ip': '10.0.0.1/8'}
Getting Hosts and Switches.
Getting controller selection:ref
Getting controller selection:ref
Getting Links.
*** Configuring hosts
h1 h2
**** Starting 2 controllers
c0 c1
**** Starting 1 switches
s1
No NetFlow targets specified.
No sFlow targets specified.
New Prefs = None
*** Stopping 2 controllers
c0 c1
*** Stopping 4 terms
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
New Prefs = None
Getting Hosts and Switches.
Getting controller selection:ref
Getting controller selection:ref
Getting Links.
*** Configuring hosts
h1 h2
**** Starting 2 controllers
c0 c1
**** Starting 1 switches
s1
No NetFlow targets specified.
No sFlow targets specified.

```

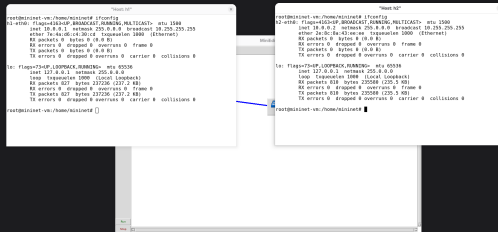
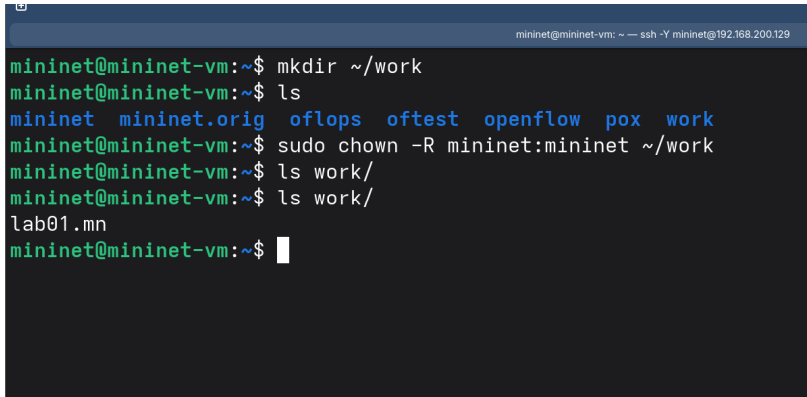


Рисунок 22: Проверка автоматически назначенных IP-адресов хостов h1 и h2

Сохранение топологии

- Подготовлен рабочий каталог ~/work.
- Топология сохранена в файл lab01.mn.

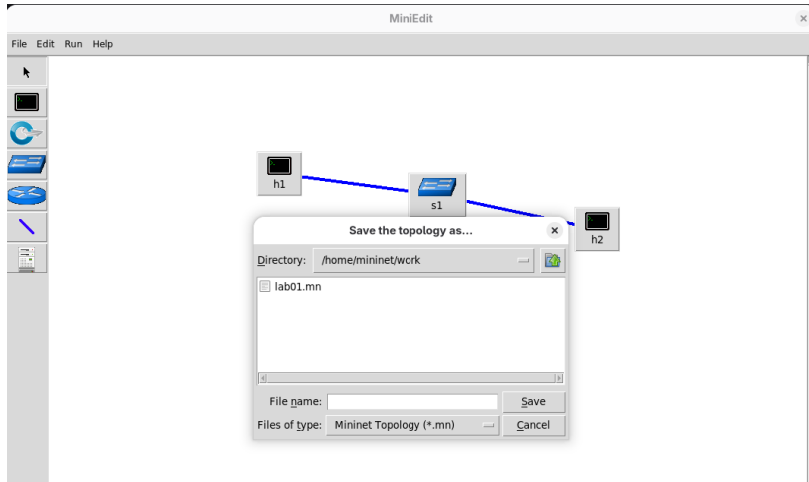


```
mininet@mininet-vm: ~ -- ssh -Y mininet@192.168.200.129
mininet@mininet-vm:~$ mkdir ~/work
mininet@mininet-vm:~$ ls
mininet  mininet.orig  oflops  oftest  openflow  pox  work
mininet@mininet-vm:~$ sudo chown -R mininet:mininet ~/work
mininet@mininet-vm:~$ ls work/
mininet@mininet-vm:~$ ls work/
lab01.mn
mininet@mininet-vm:~$
```

Рисунок 23: Проверка сохранённого файла топологии lab01.mn в рабочем каталоге

Открытие сохранённой топологии

- В MiniEdit открыт каталог `/home/mininet/work`.
- Выбран сохранённый файл `lab01.mn`.



Проверка загруженной топологии

- Файл lab01.mn успешно загружен.
- Восстановлены хосты h1 и h2.
- Восстановлен коммутатор s1.
- Восстановлены соединения между узлами.

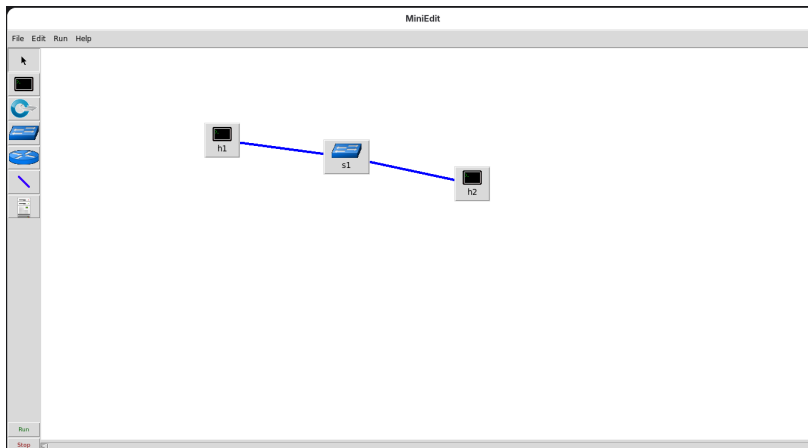


Рисунок 25: Загруженная из файла lab01.mn топология сети в MiniEdit

Раздел 4

Заключение

- Подготовлена среда Mininet.
- Создана топология из двух хостов и одного коммутатора.
- Проверены сетевые интерфейсы и связность хостов.
- Создана аналогичная топология в MiniEdit.
- IP-адреса настроены вручную и через IP Base.
- Топология сохранена в файл lab01 .mn.
- Проверена повторная загрузка сохранённой топологии.